

0 4 1

Put a tick (✓) in the box next to the process that occurs in anaerobic respiration but does **not** occur in aerobic respiration.

[1 mark]

Phosphorylation of glucose

☐

Reduction of NAD

☐

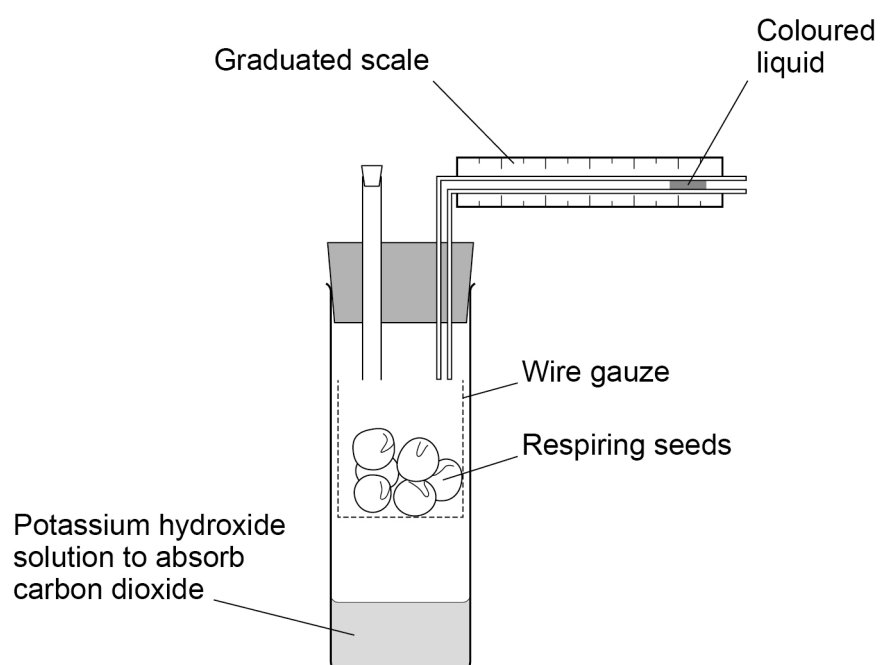
Reduction of pyruvate

☐

Substrate-level phosphorylation

☐

A student used the apparatus shown in **Figure 1** to measure the rate of aerobic respiration of seeds for 48 hours.

Figure 1

0	4	.	2
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During the 48 hours, the coloured liquid moved to the left.

Explain why.

[3 marks]

0	4	.	3
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Apart from time, give **two** measurements the student would have to make to determine the rate of aerobic respiration of these seeds in $\text{cm}^3 \text{ hour}^{-1}$

[2 marks]

1

2

Question 4 continues on the next page

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0 4 . 4

The student used the same apparatus to determine the volume of carbon dioxide the seeds produced during 48 hours.

Give the change the student would need to make to the contents of the apparatus **and** describe how he could calculate the volume of carbon dioxide produced.

[3 marks]

0 4 . 5

The student calculated that during the 48 hours, $6.2 \times 10^{-4} \text{ cm}^3$ of oxygen was absorbed by 40 g of seeds.

Calculate the oxygen uptake in $\text{cm}^3 \text{ g}^{-1} \text{ hour}^{-1}$

[1 mark]

Answer _____ $\text{cm}^3 \text{ g}^{-1} \text{ hour}^{-1}$

Turn over for the next question

*Do not
write*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

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